



Prospective study of beginner running groups: psychological predictors and outcomes of participation

Carolyn R. Plateau, Justine Anthony, Stacy A. Clemes & Clare D. Stevinson

To cite this article: Carolyn R. Plateau, Justine Anthony, Stacy A. Clemes & Clare D. Stevinson (2022): Prospective study of beginner running groups: psychological predictors and outcomes of participation, Behavioral Medicine, DOI: [10.1080/08964289.2022.2100865](https://doi.org/10.1080/08964289.2022.2100865)

To link to this article: <https://doi.org/10.1080/08964289.2022.2100865>



© 2022 The Author(s). Published with license by Taylor & Francis Group, LLC



Published online: 12 Aug 2022.



Submit your article to this journal [↗](#)



Article views: 319



View related articles [↗](#)



View Crossmark data [↗](#)

Prospective study of beginner running groups: psychological predictors and outcomes of participation

Carolyn R. Plateau^{a,b} , Justine Anthony^a , Stacy A. Clemes^{a,b,c}  and Clare D. Stevinson^{a,b} 

^aSchool of Sport, Exercise and Health Sciences, Loughborough University, Loughborough, United Kingdom; ^bNational Centre for Sport and Exercise Medicine East Midlands, Loughborough University, Loughborough, United Kingdom; ^cNational Institute for Health Research (NIHR) Leicester Biomedical Research Centre, University Hospitals of Leicester NHS Trust and University of Leicester, Leicester, United Kingdom

ABSTRACT

Running is an example of vigorous activity that leads to important health benefits if maintained. Beginner running groups provide supportive training programs to help people progress from walking to sustained running. This study explored the characteristics of individuals joining beginner running groups and the outcomes they achieve. New members of beginner running groups ($n=141$; mean age 43 years, 122 female) completed online assessments at the start of their group program with 63 participants (45%) also completing a follow-up assessment at the end of the program. Validated scales were used to assess exercise behavior, mental wellbeing, self-efficacy, running identity and social physique anxiety. The majority of participants had low exercise levels at the start of the program (63%, $n=89$). By the program end, 47 participants (75% of those completing the follow-up assessment) reported meeting the training goal (running for 30 minutes continuously) with self-efficacy, program adherence and younger age representing significant predictors of success. Significant improvements in exercise levels, mental wellbeing, self-efficacy, running identity and social physique anxiety were observed by the end of the program. In conclusion, beginner running programs attract low active individuals and may lead to improved levels of exercise and psychological outcomes. Additional research is needed to examine the extent to which improvements are sustained longer term.

ARTICLE HISTORY

Received 26 February 2022
Revised 26 May 2022
Accepted 27 June 2022

KEYWORDS

adherence; running; self-efficacy; social physique anxiety; wellbeing

Introduction

Public health guidelines recommend that adults achieve 150 minutes of moderate intensity physical activity or 75 minutes of vigorous activity weekly. Running is one example of vigorous physical activity suggested within the guidelines.¹ The potential population health benefits of running are considerable with a recent meta-analysis demonstrating a 23-30% reduction in cardiovascular, cancer and all-cause mortality, regardless of running volume.² The authors suggested that increasing rates of participation in running would lead to substantial improvements in population longevity and quality of life. The benefits of running also extend to psychological wellbeing with an extensive review of evidence indicating that running of all intensities can improve mood and mental health.³

Despite the benefits, taking up running can be difficult in adulthood. Commonly perceived barriers include aging, excess body weight and low confidence

in running ability.⁴ Beginner running groups are aimed at developing the fitness and confidence to run among novices with support provided from the group leader and other participants. Being part of a group is particularly valued among recreational runners for increasing enjoyment and persistence.⁵⁻⁷ Beginner programs typically involve walk-run intervals where the amount of running is incrementally increased over an 8-12-week period with the goal of running continuously for a minimum distance or time (e.g., 5 kilometers or 30 minutes).

There has been some preliminary research with members of beginner runner groups to understand the characteristics of those who join such programs. In the Netherlands, a nationwide 6-week 'Start to Run' program is organized by the Dutch Athletics Federation twice yearly and attracts over 8,000 members. Profiles of two separate samples of participants indicated that they were mostly female, middle-aged, overweight and new to running.^{8,9} Although completion rates of the

CONTACT Clare D. Stevinson  C.D.Stevinson@lboro.ac.uk  School of Sport, Exercise and Health Sciences, Loughborough University, Loughborough LE11 3TU, United Kingdom

© 2022 The Author(s). Published with license by Taylor & Francis Group, LLC

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (<http://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way.

program were not documented, available follow-up data suggested that most participants continued to run⁹ or meet physical activity guidelines¹⁰ at 6 months. Discontinuation of running was associated with being female and lacking recent experience of running, with injury being the most common reason cited.⁹

In addition to long-term impact, there is a need for research to examine predictors of program adherence and potential outcomes of completing beginner runner training. Through understanding who is most likely to join, complete, and benefit from, programs, there is scope for widening participation and improving delivery to ensure maximum public health value of these initiatives.

Research on social cognitive theory points toward the importance of self-efficacy in initiating and maintaining physical activity.¹¹ Qualitative studies have identified low confidence among new runners⁴ and the development of self-confidence through meeting running goals.⁷ Additional data from quantitative research could indicate the extent that self-efficacy for running represents a determinant or outcome of participation in beginner programs. Another psychological construct that merits attention is role identity. Studies of running group members have shown how identity as a runner can develop through group participation¹² and is vital for maintenance of independent running outside of the group environment.¹³ Similarly, anxiety around appearance, or fear of judgment of one's body by others whilst exercising, can deter exercise engagement.^{14,15} Further research on social physique anxiety in beginner runners will help understand if it influences participation and whether it might change over time.

In summary, beginner running programmes have important public health potential in supporting engagement with vigorous physical activity. However, there is limited understanding of the characteristics of individuals who take part, factors that influence their completion of such programmes and the outcomes that can be achieved from participation. This study therefore had two main aims. First, the sociodemographic, physical and psychological characteristics of participants of group-based beginner running programs were explored and used to identify predictors of successful completion. Second, changes in these variables between starting and finishing training programs were examined to indicate outcomes that can be achieved through taking part.

Methods

Participants

Organizers of group-based beginner running programs were identified via a national website (RunTogether)

listing groups throughout the United Kingdom (UK), or specific group Facebook pages. Group leaders were invited to share information about the study with their new members if they were leading a beginner level training program for adults that would be commencing in the next four months. Individuals could pre-register to join programs, or in some cases, simply turn up to the first session.

Group programs were led by two or more qualified run leaders in community settings with the typical group size ranging from 10 to 25. Run leaders held the England Athletics Leadership in Running Fitness qualification. Programs were either based directly on the 9-week Couch to 5k training principles¹⁶ or a similar format that involved short run repeats with walk recovery periods (e.g. 6 × 1-minute run, 2-minute walk). The time spent running was progressed each week with the goal of completing either 5 km of distance or 30 minutes of continuous running by the end of the program. Although a standard program was used for groups, leaders provided tailored support or advice for some individuals based on specific needs. Sessions typically lasted a total 45-60 minutes including time spent warming up and cooling down before and after the training. Most leaders offered one or more group sessions per week and encouraged participants to repeat sessions independently as 'homework' in order to complete three sessions each week over the 9-week training period.

Individuals aged at least 18 years who were registered to start group training, or who had completed no more than the first two weeks of the program, were invited to take part in the study. A total of 141 participants from 33 separate beginner groups completed baseline measures (86.5% female; mean age 43.13 ± 10.11 years) and 63 were retained at follow-up (44.6% retention rate).

Procedure

The study had a prospective design with participant questionnaires completed at baseline and after program completion (10 weeks after baseline) via an online platform. The study was approved by the Ethics Advisory Committee for Human Participants at Loughborough University (C18-08) and all participants provided consent online after reading full study information, which enabled access to the baseline questionnaire. Entry into a prize draw to win one of four £25 shopping vouchers was offered for completing both assessments. Non-responders at the second assessment point were sent email reminders after one week.

Measures

Participants reported sociodemographic information including age, sex, marital status, educational level, employment status, ethnic background and limiting disabilities. They also self-reported height and weight to enable estimates of body mass index (BMI).

Self-reported exercise behavior was assessed with the Godin Leisure-Time Exercise Questionnaire.¹⁷ Participants were asked to indicate their weekly frequency of mild, moderate, and strenuous intensity activity undertaken in their free time. The average duration of sessions was also reported to enable the volume of exercise to be estimated. The reliability and validity of the instrument compares favorably to other self-report measures of physical activity¹⁸ and objective measures of fitness¹⁹.

Validated scales were used for measuring four psychological variables. Running identity was assessed through the 2-item self-identity subscale of the Running Identity Scale.²⁰ In its initial development the measure was found to successfully discriminate between professional, collegiate, serious, recreational and non-runners and had good internal consistency.²⁰ Self-efficacy was assessed with the 9-item Self-Efficacy for Exercise scale²¹ which examines participant confidence in their ability to exercise in different circumstances (e.g., when tired or stressed). For the purposes of this study, the word 'exercise' was replaced with 'running'. A Cronbach alpha of .85 was calculated for the current sample, which is similar to studies using the original measure.²¹ The 7-item version of the Social Physique Anxiety Scale²² was used for assessing anxiety around physical appearance. The scale has shown a good level of test-retest reliability²³ and construct validity²² in relation to other measures of body consciousness and had high internal consistency in the current study (Cronbach alpha of .93). Finally, mental wellbeing was assessed with the 6-item Short Depression Happiness Scale.²⁴ The measure has demonstrated acceptable test-retest reliability and internal consistency as well as strong convergent validity with other depression scales in US and UK samples²⁴. Acceptable internal consistency was shown in the current study (Cronbach alpha of .80). All study measures are widely used for research in UK samples.

At follow-up participants self-reported several program outcomes including adherence (the percentage of running sessions they attended during the program), and if they had experienced any injury that affected participation. They also reported program success (whether they had succeeded in being able to run for 30 minutes continuously by the end of the program) and if they intended to continue running beyond the completion of the beginner program.

Data analysis

All statistical analyses were conducted using IBM SPSS Statistics version 24. Descriptive statistics (frequency, mean, standard deviation) were then calculated for participants who completed questionnaires at baseline ($n=141$) and for those who completed baseline and follow-up ($n=63$) and differences between the samples were examined.

Longitudinal analyses were conducted with the 63 participants who completed both assessments. Most variables were not normally distributed, so non-parametric tests were employed for correlational analyses and examining changes pre- to post-program. Spearman's correlations were used to determine whether baseline variables were associated with program adherence (percentage of sessions completed). Exploratory binary logistic regression analysis was conducted to explore predictors of program success (i.e., being able to run for 30 minutes continuously by the end of the program). Baseline demographic (age), physical (BMI; exercise) and psychological (running identity, self-efficacy, social physique anxiety, wellbeing) variables were included in the model along with program adherence and injury. Finally, Wilcoxon tests of difference were employed to determine changes in physical and psychological variables from baseline to follow-up with Cohen's d_z effect sizes calculated for repeated measures designs.²⁵

Results

Participant characteristics

Table 1 summarizes the baseline characteristics for all participants ($n=141$) and for those who completed both assessment points for the study ($n=63$). Study completers had a significantly lower BMI than those who were lost to follow-up ($26.73 \pm 4.85 \text{ kg/m}^2$ versus $29.62 \pm 6.03 \text{ kg/m}^2$; $Z=2.98$, $p=.003$; $r=.23$). No other significant differences were observed between study completers and non-completers. Similarly, BMI was the only independent predictor of study attrition from a logistic regression analysis ($OR=1.14$, $95\% \text{ CI}=1.04$ to 1.24 , $p=.004$). Participants ranged in age from 18 to 68 years (mean 43.13 ± 10.11), with high proportions of the sample being female (86.5%, $n=122$), employed (85.1%, $n=120$), married or cohabiting (75.2%, $n=106$) and of white ethnicity (91.5%, $n=129$). Overall exercise levels were low with the mean number of weekly minutes of moderate to vigorous activity (114.5 minutes) being below minimum recommended levels for health benefit (150 minutes of moderate intensity or 75 minutes of vigorous intensity activity or an equivalent combination). Specifically, 89 participants (63.1%) were

Table 1. Participant baseline characteristics.

	Baseline sample (n = 141)	Study completers (n = 63)
Age in years (M ± SD)	43.13 ± 10.11	43.68 ± 10.77
Female	122 (86.5%)	56 (88.9%)
White ethnic group	129 (91.5%)	60 (95.2%)
In current employment	120 (85.1%)	52 (82.5%)
Completion of higher education	79 (56.0%)	38 (60.3%)
Married/cohabiting	106 (75.2%)	48 (76.2%)
Limiting disability	34 (24.1%)	18 (28.6%)
Overweight/obese	91 (64.5%)	36 (57.1%)
BMI in kg/m ² (M ± SD)	28.37 ± 5.72	26.73 ± 4.69
Moderate/vigorous exercise in min/w (M ± SD)	114.50 ± 191.76	88.65 ± 109.99
Identified as current non-runner	118 (83.7%)	52 (82.5%)
Running identity score (M ± SD)	7.2 ± 3.0	7.1 ± 2.8
Self-efficacy score (M ± SD)	43.0 ± 17.6	41.9 ± 16.2
Social physique anxiety score (M ± SD)	3.2 ± 1.1	3.01 ± 1.1
Mental wellbeing score (M ± SD)	12.3 ± 3.4	12.77 ± 3.4

Key: BMI: body mass index; kg: kilogram; m: meter; min: minute; w: week.

below this threshold and 56 (39.7%) reported performing no moderate-vigorous activity at all.

Program outcomes

At the follow-up assessment, the overall adherence rate (defined as the percentage of running sessions reported as attended during the program) was 69.6 ± 31.0%. The only baseline variable to be associated with program adherence was BMI ($r_s = -.35$, $p = .007$). A total of 21 participants (33.3%) reported experiencing an injury during the training program with eight citing the injury as preventing program completion. Fifty-three participants (84.1%) intended to continue running beyond the end of the program.

Forty-seven participants (74.6% of those completing follow-up assessments) reported having successfully met the program goal of being able to run continuously for 30 minutes. The logistic regression model was statistically significant ($\chi^2 (9) = 36.37$, $p < .001$) and represented a good fit to the data ($\chi^2 (8) = 3.05$, $p = .93$). The model explained 66% of the variance in program success. Age, baseline self-efficacy and adherence were statistically significant predictors of program success (Table 2). Increasing age was associated with lower odds of reaching the program goal (OR = 0.87, 95% CI = 0.77 to 0.98, $p = .018$). Higher program adherence was related to increased likelihood of a successful outcome (OR = 1.04, 95% CI = 1.00 to 1.08, $p = .039$). Finally, greater baseline self-efficacy for running was linked to an increased likelihood of program success (OR = 1.21, 95% CI = 1.06 to 1.37, $p = .004$).

Table 3 shows changes in psychological and physical outcomes over the course of the study. Significant increases were observed for self-reported moderate to vigorous exercise from baseline to follow up. Significant increases in scores were also observed for running

Table 2. Logistic regression analysis of predictors of program success (reaching the goal of 30 minutes continuous running).

	B	SE	Wald	<i>p</i>	Odds ratio	95% CI
Age	−0.145	0.061	5.585	0.018	0.87	0.77–0.98
BMI	−0.195	0.142	1.884	0.170	0.82	0.62–1.09
Moderate/vigorous exercise	−0.004	0.004	1.140	0.286	1.00	0.99–1.00
Running identity	−0.033	0.198	0.027	0.869	0.97	0.66–1.43
Self-efficacy	0.187	0.064	8.440	0.004	1.21	1.06–1.37
Social physique anxiety	−0.191	0.545	0.123	0.726	0.83	0.28–2.24
Mental wellbeing	−0.119	0.173	0.474	0.491	0.89	0.63–1.20
Adherence	0.041	0.020	4.269	0.039	1.04	1.00–1.08
Injury	1.330	1.011	1.729	0.189	3.78	0.52–27.4
Intercept	6.236	6.912	0.814	0.367	–	–

Table 3. Change in self-reported outcomes from baseline to follow up (n = 63).

	Baseline M ± SD	Follow up M ± SD	Z	P	<i>d_z</i>
BMI in kg/m ²	26.73 ± 4.69	26.75 ± 4.94	−1.48	0.138	0.01
Moderate/vigorous exercise in min/w	88.65 ± 110.00	161.27 ± 147.41	−4.30	<0.001	0.58
Running identity	7.1 ± 2.8	9.6 ± 2.9	−4.75	<0.001	0.73
Self-efficacy	41.9 ± 16.3	45.4 ± 17.0	−2.28	0.022	0.19
Social physique anxiety	3.0 ± 1.1	2.6 ± 0.9	−4.59	<0.001	0.58
Mental wellbeing	12.8 ± 3.4	13.9 ± 3.9	−2.92	0.004	0.38

Key: BMI: body mass index; kg: kilogram; m: meter; min: minute; w: week.

identity, self-efficacy and psychological wellbeing, while social physique anxiety significantly decreased. No changes were observed for BMI from baseline to follow up.

Discussion

This study explored the characteristics of beginner runner group members, identified predictors of training success, and examined changes in outcomes at the end of the program. The profiling data indicated that participants were typically female, overweight, low active and new to running. Approximately three-quarters of those completing the study reported meeting the program goal (i.e., being able to run for 30 minutes continuously), with younger age, higher adherence to training and greater self-efficacy identified as significant predictors. Finally, small significant improvements in exercise, mental wellbeing, self-efficacy, running identity and social physique anxiety were recorded at the end of the program.

The characteristics of beginner runners from UK groups examined here are in line with those reported from similar schemes in the Netherlands.^{8,9} It demonstrates the suitability of such programs for attracting population groups traditionally lower in physical activity (i.e., females, middle-aged, overweight). However, it was notable that the sample was dominated by those

with high education levels and white ethnic backgrounds, which are consistently associated with participation disparities in leisure-time physical activity.²⁶ Cultural and social concerns have been identified as significant exercise barriers among black and minority ethnic groups²⁷ and the need to address these within community exercise opportunities is critical in the drive to reduce inequalities in engagement.²⁸

In addition to sociodemographic data, this study provided insight into the psychological profile of beginner runners. Without prior experience of running, it is unsurprising that identity for running and self-efficacy for running were both low. Scores on the self-identity sub-scale of the Running Identity Scale were below those recorded for samples of both serious/collegiate/professional competitors and recreational runners.²⁰ Similarly, self-efficacy for running scores were lower than those reported with the Self-Efficacy for Exercise scale for samples of community dwelling older adults.^{29,30} Meanwhile, social physique anxiety scores among beginner runners were notably higher than those observed among regular exercisers recruited from fitness clubs.³¹ These psychological variables have been highlighted as particularly important in understanding physical activity participation among women,³² who represented a large proportion of the sample in the current study.

Self-reported adherence to the training program was 69.6% which compares favorably with other studies of beginner running interventions where adherence rates between 45 and 65% were recorded.^{33–35} Adherence was inversely associated with baseline BMI. This may suggest that heavier participants find the training program more challenging to complete and that adaptations to the typical 9-week Couch to 5k program might be appropriate for some individuals. One example of this being addressed was in a start-to-run intervention designed for obese adults in Denmark.³⁴ Comparisons of two volumes of weekly running (3km and 6km) confirmed that the lower volume program resulted in higher adherence and fewer injuries among participants.

Injury was reported by one in three participants of the current study although it is possible that this is an underestimate given the substantial loss of follow-up for which the reasons are unknown. Injury is recognized to be common among experienced and novice runners³⁶ and an injury rate of 49% was reported for new runners undertaking structured programs such as Couch to 5k.³⁷ This highlights the potential need to include education around injury prevention for both group leaders and participants of beginner running programs. Recent research has been

directed at developing and evaluating online delivery of evidence-based injury prevention education aimed at novice and experienced runners.^{38–40}

A high proportion of study completers reported successfully reaching the program goal (i.e., ability to run for 30 minutes continuously) with adherence being one of the main determinants. Baseline self-efficacy was also an important predictor of success which is consistent with existing research identifying confidence as pivotal for exercise engagement.¹¹ Self-efficacy for exercise can be developed through perceptions of success, appropriate social comparisons and positive feedback.⁴¹ Group-based programs offer the opportunity for all of these through peer support and group leadership.⁴² In addition, testimonials or visits from successful completers of programs may be further ways of promoting self-efficacy among current participants based on promising evidence for narrative interventions⁴³ and vicarious experiences.⁴⁴

Self-efficacy was one of the psychological outcomes that was significantly improved by the end of the training program along with running identity, social physique anxiety and mental wellbeing. The increase in running identity reached a level demonstrated for recreational runners,²⁰ and based on previous evidence, raises the likelihood of continued independent running outside of the group setting.¹³ The reduction in social physique anxiety is notable given the barrier it represents to physical activity engagement.⁴⁵ Self-reported moderate-vigorous exercise increased by more than 70 minutes per week and, if maintained long-term, would translate into important fitness and health benefits.⁴⁶ Completion of the training sessions account for some of this increase, but it appears that additional exercise has been undertaken facilitated by the higher levels of running identity and self-efficacy. Qualitative data from a sub-set of the study sample ($n=34$) provided some insight into the psychological changes observed.⁴⁷ Participants gained confidence and pride in their ability to run as they became aware of their improving fitness level over time through reduced breathlessness and fatigue. They developed a sense of achievement as program milestones were reached and attributed their ability to persist with the training to the support received from group leaders and members.

While this study points toward some significant benefits for participants of beginner running programs over a relatively short time frame, several limitations should be noted. The observational design meant that no control arm existed for accurately quantifying the degree of change in exercise and psychological outcomes. Furthermore, there was significant attrition

with only 45% of participants completing both assessments. It is not known whether participants who did not complete the study also discontinued their beginner running program and, if so, the reasons for drop-out. Similarly, information on the number of group participants invited to the study and those who chose not to join the study are unavailable. Although the number achieving the program goal was high among those who continued with the study, it is important to acknowledge that outcomes may be inflated by attrition bias. Further evaluation is needed to understand if this is a true reflection of the success rate of these programs and if running behavior and other outcomes are sustained in the long-term.

This study exclusively focused on group-based programs and does not provide data on individuals following beginner running programs independently. Digital tools (e.g., the Couch to 5k smartphone application) are popular and offer greater flexibility for completion than scheduled group sessions. Research is needed to explore any differences in the characteristics of participants who use digital tools instead of face-to-face groups, and whether outcomes vary by delivery mode. Finally, it is noted that all data were self-reported and more studies with objective measures of exercise behavior and fitness indices will help improve understanding of the outcomes that can be achieved.

Conclusions

In conclusion, these results suggest that group-based beginner running programs are attracting low-active and overweight individuals who would benefit from vigorous physical activity and that participation leads to increased exercise behavior and intentions to continue running. Self-efficacy at the start of the program may be an important determinant of success and can also be developed during the program along with positive changes in social physique anxiety and mental wellbeing. Promoting engagement with beginner running programs among more diverse population groups is essential to ensure that the wider public health potential of these programs can be realized.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Sport and Exercise Beacon at Loughborough University. Stacy Clemes is supported by the NIHR Leicester Biomedical Research Center. The views

expressed are those of the authors and not necessarily those of the NHS, the NIHR or the Department of Health and Social Care.

ORCID

Carolyn R. Plateau  <http://orcid.org/0000-0002-1256-1353>
 Justine Anthony  <http://orcid.org/0000-0002-1899-1100>
 Stacy A. Clemes  <http://orcid.org/0000-0001-5612-5898>
 Clare D. Stevinson  <http://orcid.org/0000-0002-7552-1412>

References

- [1] Department of Health and Social Care. UK Chief Medical Officers' physical activity guidelines; 2019. <https://www.gov.uk/government/publications/physical-activity-guidelines-uk-chief-medical-officers-report>.
- [2] Pedisic Z, Shrestha N, Kovalchik S, et al. Is running associated with a lower risk of all-cause, cardiovascular and cancer mortality, and is the more the better? A systematic review and meta-analysis. *Br J Sports Med*. 2020;54(15):898–905. doi:10.1136/bjsports-2018-100493.
- [3] Oswald F, Campbell J, Williamson C, Richards J, Kelly P. A scoping review of the relationship between running and mental health. *IJERPH*. 2020;17(21):8059. doi:10.3390/ijerph17218059.
- [4] Griffin M, Phoenix C. Becoming a runner: Big, middle and small stories about physical activity participation in later life. *Sport Educ Soc*. 2016;21(1):11–27. doi:10.1080/13573322.2015.1066770.
- [5] Carnes AJ, Mahoney SE. Cohesion is associated with perceived exertion and enjoyment during group running. *J Exer Physiol Online*. 2016;19(6):24–39.
- [6] Polikowsky L. Factors affecting adherence to an 18-week running program for women [Masters of Science Thesis]. University of New Mexico; 2012. <https://conservancy.umn.edu/handle/11299/132422>.
- [7] Rothberger SM. A qualitative examination of hope and physical activity in the No Limits running program. *Int J Kinesiol Higher Educ*. 2018;2(2):54–65. doi:10.1080/24711616.2017.1403871.
- [8] Kluitenberg B, van der Worp H, Huisstede BM, et al. The NLstart2run study: Training-related factors associated with running-related injuries in novice runners. *J Sci Med Sport*. 2016;19(8):642–646. doi:10.1016/j.jsams.2015.09.006.
- [9] Fokkema T, Hartgens F, Kluitenberg B, et al. Reasons and predictors of discontinuation of running after a running program for novice runners. *J Sci Med Sport*. 2019;22(1):106–111. doi:10.1016/j.jsams.2018.06.003.
- [10] Ooms L, Veenhof C, de Bakker DH. Effectiveness of Start to Run, a 6-week training program for novice runners, on increasing health-enhancing physical activity: a controlled study. *BMC Public Health*. 2013;13:697. doi:10.1186/1471-2458-13-697.
- [11] Young MD, Plotnikoff RC, Collins CE, Callister R, Morgan PJ. Social cognitive theory and physical activity: a systematic review and meta-analysis. *Obes Rev*. 2014;15(12):983–995. doi:10.1111/obr.12225.

- [12] Evans MB, McLaren C, Budziszewski R, Gilchrist J. When a sense of “we” shapes the sense of “me”: exploring how groups impact running identity and behaviour. *Self Identity*. 2019;18(3):227–246. doi:10.1080/15298868.2018.1436084.
- [13] Strachan SM, Shields CA, Glassford A, Beatty J. Role and group identity and adjustment to the possibility of running group disbandment. *Psychol Sport Exer*. 2012;13(4):436–443. doi:10.1016/j.psychsport.2012.01.007.
- [14] Brunet J, Sabiston C. Social physique anxiety and physical activity: A self-determination theory perspective. *Psychol Sport Exer*. 2009;10(3):329–335. doi:10.1016/j.psychsport.2008.11.002.
- [15] Mensinger JL, Meadows A. Internalized weight stigma mediates and moderates physical activity outcomes during a healthy living program for women with high body mass index. *Psychol Sport Exer*. 2017;30:64–72. doi:10.1016/j.psychsport.2017.01.010.
- [16] National Health Service. Get running with couch to 5K; 2017. <https://www.nhs.uk/live-well/exercise/get-running-withcouch-to-5k/>.
- [17] Godin G, Shephard RJ. A simple method to assess exercise behaviour in the community. *Can J Appl Sport Sci*. 1985;10(3):141–146.
- [18] Jacobs DRJ, Ainsworth BE, Hartman TJ, Leon AS. A simultaneous evaluation of 10 commonly used physical activity questionnaires. *Med Sci Sport Exer*. 1993;25:81–91.
- [19] Amireault S, Godin G. The Godin-Shephard leisure-time physical activity questionnaire: validity evidence supporting its use for classifying healthy adults into active and insufficiently active categories. *Percept Mot Skills*. 2015;120(2):604–622. doi:10.2466/03.27.PMS.120v19x7.
- [20] Bejar MP. Development of the Running Identity Scale; 2016. Unpublished Manuscript. https://trace.tennessee.edu/utk_graddiss/3679/.
- [21] Resnick B, Jenkins LS. Testing the reliability and validity of the self-efficacy for exercise scale. *Nurs Res*. 2000;49(3):154–159. doi:10.1097/00006199-200005000-00007.
- [22] Motl RW, Conroy DE. Validity and factorial invariance of the social physique anxiety scale. *Med Sci Sport Exer*. 2000;32(5):1007–1017.
- [23] Scott LA, Burke KL, Joyner AB, Brand JS. Examining the stability of the 7-item Social Physique Anxiety Scale using a test-retest method. *Meas Phys Educ Exerc Sci*. 2004;8(2):57–62. doi:10.1207/s15327841mpee0802_1.
- [24] Joseph S, Linley PA, Harwood J, Lewis CA, McCollam P. Rapid assessment of well-being: The Short Depression-Happiness Scale (SDHS). *Psychol Psychother*. 2004;77(Pt 4):463–478. doi:10.1348/1476083042555406.
- [25] Lakens D. Calculating and reporting effect sizes to facilitate cumulative science: a practical primer for t-tests and ANOVAs. *Front Psychol*. 2013;4:863.
- [26] Scholes S, Bann D. Education-related disparities in reported physical activity during leisure-time, active transportation, and work among US adults: repeated cross-sectional analysis from the National Health and Nutrition Examination Surveys, 2007 to 2016. *BMC Public Health*. 2018;18(1):926. doi:10.1186/s12889-018-5857-z.
- [27] Ige-Elegbede J, Pilkington P, Gray S, Powell J. Barriers and facilitators of physical activity among adults and older adults from Black and Minority Ethnic groups in the UK: A systematic review of qualitative studies. *Prev Med Rep*. 2019;15:100952. doi:10.1016/j.pmedr.2019.100952.
- [28] Hämäläinen RM, Sandu P, Syed AM, Jakobsen MW. An evaluation of equity and equality in physical activity policies in four European countries. *Int J Equity Health*. 2016;15(1):191. doi:10.1186/s12939-016-0481-y.
- [29] Perkins JM, Multhaup KS, Perkins HW, Barton C. Self-efficacy and participation in physical and social activity among older adults in Spain and the United States. *Gerontologist*. 2008;48(1):51–58. doi:10.1093/geront/48.1.51.
- [30] Miller KJ, Mesagno C, McLaren S, Grace F, Yates M, Gomez R. Exercise, mood, self-efficacy, and social support as predictors of depressive symptoms in older adults: direct and interaction effects. *Front Psychol*. 2019;10(2145):2145.
- [31] Thøgersen-Ntoumani C, Ntoumanis N. The role of self-determined motivation in the understanding of exercise-related behaviours, cognitions and physical self-evaluations. *J Sports Sci*. 2006;24(4):393–404. doi:10.1080/02640410500131670.
- [32] Edwards ES, Sackett SC. Psychosocial variables related to why women are less active than men and related health implications. *Clin Med Insights Womens Health*. 2016;9(Suppl 1):47–56.
- [33] Buist I, Bredeweg SW, van Mechelen W, Lemmink KA, Pepping GJ, Diercks RL. No effect of a graded training program on the number of running-related injuries in novice runners: a randomized controlled trial. *Am J Sports Med*. 2008;36(1):33–39. doi:10.1177/0363546507307505.
- [34] Bertelsen ML, Hansen M, Rasmussen S, Nielsen RO. the start-to-run distance and running-related injury among obese novice runners: a randomized trial. *Int J Sports Phys Ther*. 2018;13(6):943–955. doi:10.26603/ijsp20180943.
- [35] Looyestyn J, Kernot J, Boshoff K, Maher C. A web-based, social networking beginners’ running intervention for adults aged 18 to 50 years delivered via a Facebook group: randomized controlled trial. *J Med Internet Res*. 2018;20(2):e67. doi:10.2196/jmir.7862.
- [36] Kluitenberg B, van Middelkoop M, Diercks R, van der Worp H. What are the differences in injury proportions between different populations of runners? a systematic review and meta-analysis. *Sports Med*. 2015;45(8):1143–1161. doi:10.1007/s40279-015-0331-x.
- [37] Linton L, Valentin S. Running with injury: A study of UK novice and recreational runners and factors associated with running related injury. *J Sci Med Sport*. 2018;21(12):1221–1225. doi:10.1016/j.jsams.2018.05.021.
- [38] Fokkema T, de Vos RJ, van Ochten JM, et al. Preventing running-related injuries using evidence-based online advice: the design of a randomised-controlled trial. *BMJ Open Sport Exerc Med*. 2017;3(1):e000265. doi:10.1136/bmjsem-2017-000265.
- [39] Fokkema T, de Vos R-J, van Ochten JM, et al. Online multifactorial prevention programme has no effect on

- the number of running-related injuries: a randomised controlled trial. *Br J Sports Med.* 2019;53(23):1479–1485. doi:[10.1136/bjsports-2018-099744](https://doi.org/10.1136/bjsports-2018-099744).
- [40] Fokkema T, de Vos RJ, Visser E, et al. Enhanced injury prevention programme for recreational runners (the SPRINT study): design of a randomised controlled trial. *BMJ Open Sport Exerc Med.* 2020;6(1):e000780. doi:[10.1136/bmjsem-2020-000780](https://doi.org/10.1136/bmjsem-2020-000780).
- [41] Samson A, Solmon M. Examining the sources of self-efficacy for physical activity within the sport and exercise domains. *Int Rev Sport Exer Psychol.* 2011;4(1):70–89. doi:[10.1080/1750984X.2011.564643](https://doi.org/10.1080/1750984X.2011.564643).
- [42] Martin Ginis KA, Nigg CA, Smith AL. Peer-delivered physical activity interventions: An overlooked opportunity for physical activity promotion. *Transl Behav Med.* 2013;3(4):434–443. doi:[10.1007/s13142-013-0215-2](https://doi.org/10.1007/s13142-013-0215-2).
- [43] Perrier MJ, Martin Ginis KA. Changing health promoting behaviours through narrative interventions: A systematic review. *J Health Psychol.* 2018;23(11):1499–1517. doi:[10.1177/1359105316656243](https://doi.org/10.1177/1359105316656243).
- [44] Ashford S, Edmunds J, French DP. What is the best way to change self-efficacy to promote lifestyle and recreational physical activity? A systematic review with meta-analysis. *Br J Health Psychol.* 2010;15(2):265–288. doi:[10.1348/135910709X461752](https://doi.org/10.1348/135910709X461752).
- [45] Atalay AA, Gençöz T. Critical factors of social physique anxiety: exercising and body image satisfaction. *Behav Change.* 2008;25(3):178–188. doi:[10.1375/bech.25.3.178](https://doi.org/10.1375/bech.25.3.178).
- [46] Hespanhol Junior LC, Pillay JD, van Mechelen W, Verhagen E. Meta-analyses of the effects of habitual running on indices of health in physically inactive adults. *Sports Med.* 2015;45(10):1455–1468. doi:[10.1007/s40279-015-0359-y](https://doi.org/10.1007/s40279-015-0359-y).
- [47] Stevinson C, Plateau CR, Plunkett S, et al. Adherence and health-related outcomes of beginner running programs: a 10-week observational study. *Res Q Exerc Sport.* 2022;93(1):87–95. doi:[10.1080/02701367.2020.1799916](https://doi.org/10.1080/02701367.2020.1799916).